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         NASA-GLENN CHEMICAL EQUILIBRIUM PROGRAM CEA2, FEBRUARY 5, 2004
                   BY  BONNIE MCBRIDE AND SANFORD GORDON
      REFS: NASA RP-1311, PART I, 1994 AND NASA RP-1311, PART II, 1996

 *******************************************************************************



 ### CEA analysis performed on Tue 17-Jun-2025 18:54:07

 # Problem Type: "Rocket" (Infinite Area Combustor)

 prob case = _______________5861 ro equilibrium

 # Pressure (1 value):
 p,atm= 68

 # Oxidizer/Fuel Wt. ratio (17 values):
 o/f = 2, 2.5, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10

 # You selected the following fuels and oxidizers:
 reac
 fuel H2                wt%=100.0000  t,k= 298.150
 oxid O2                wt%=100.0000  t,k= 298.150

 # Consider ONLY these selected species. (3 selected)
 only H2 HO2 O2


 # You selected these options for output:
 # short version of output
 output short
 # Proportions of any products will be expressed as Mole Fractions.
 # Heat will be expressed as siunits
 output siunits

 # Input prepared by this script:/var/www/sites/cearun/cgi-bin/CEARUN/prepareInpu
 tFile.cgi

 ### IMPORTANT:  The following line is the end of your CEA input file!
 end





              THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

           COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

 Pin =   999.3 PSIA
 CASE = _______________

             REACTANT                    WT FRACTION      ENERGY      TEMP
                                          (SEE NOTE)     KJ/KG-MOL      K
 FUEL        H2                           1.0000000        -0.000    298.150
 OXIDANT     O2                           1.0000000        -0.000    298.150

 O/F=    2.00000  %FUEL= 33.333333  R,EQ.RATIO= 3.968341  PHI,EQ.RATIO= 3.968341
```

	CHAMBER	THROAT
Pinf/P	1.0000	1.9017
P, BAR	68.901	36.232
T, K	298.13	247.45
RHO, KG/CU M	1.4929 1	9.4583 0
H, KJ/KG	0.00000	-270.71
U, KJ/KG	-461.51	-653.78
G, KJ/KG	-5924.71	-5188.34
S, KJ/(KG)(K)	19.8729	19.8729
M, (1/n)	5.371	5.371
(dLV/dLP)t	-1.00001	-1.00000
(dLV/dLT)p	0.9999	1.0000
Cp, KJ/(KG)(K)	5.3819	5.2922
GAMMAs	1.4036	1.4134
SON VEL,M/SEC	804.9	735.8
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	990.0
CF	0.7432
Ivac, M/SEC	1256.4
Isp, M/SEC	735.8

MOLE FRACTIONS

HO2	0.00005	0.00001
*H2	0.88810	0.88810
*O2	0.11185	0.11189

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 999.3 PSIA
CASE = _______________

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H2	1.0000000	-0.000	298.150
OXIDANT	O2	1.0000000	-0.000	298.150

O/F= 2.50000 %FUEL= 28.571429 R,EQ.RATIO= 3.174673 PHI,EQ.RATIO= 3.174673

	CHAMBER	THROAT
Pinf/P	1.0000	1.9014
P, BAR	68.901	36.238
T, K	298.13	247.49
RHO, KG/CU M	1.6944 1	1.0734 1

```
H, KJ/KG        -0.00000  -238.48
U, KJ/KG         -406.64  -576.07
G, KJ/KG        -5327.58 -4661.21
S, KJ/(KG)(K)    17.8702  17.8702

M, (1/n)           6.096    6.096
(dLV/dLP)t      -1.00001 -1.00000
(dLV/dLT)p       0.9999   1.0000
Cp, KJ/(KG)(K)   4.7445   4.6668
GAMMAs           1.4033   1.4129
SON VEL,M/SEC     755.4    690.6
MACH NUMBER       0.000    1.000

PERFORMANCE PARAMETERS

Ae/At                     1.0000
CSTAR, M/SEC               929.4
CF                        0.7431
Ivac, M/SEC               1179.4
Isp, M/SEC                 690.6


MOLE FRACTIONS

HO2              0.00006  0.00001
*H2              0.86393  0.86393
*O2              0.13601  0.13605

 * THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS
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             THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

          COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin =   999.3 PSIA
CASE = _______________

            REACTANT                    WT FRACTION      ENERGY      TEMP
                                         (SEE NOTE)     KJ/KG-MOL      K
FUEL        H2                           1.0000000        -0.000    298.150
OXIDANT     O2                           1.0000000        -0.000    298.150

O/F=    3.00000  %FUEL= 25.000000  R,EQ.RATIO= 2.645561  PHI,EQ.RATIO= 2.645561

                CHAMBER   THROAT
Pinf/P            1.0000   1.9011
P, BAR            68.901   36.243
T, K              298.12   247.53
RHO, KG/CU M    1.8852 1 1.1943 1
H, KJ/KG        -0.00000  -214.31
U, KJ/KG         -365.49  -517.78
G, KJ/KG        -4878.04 -4264.48
S, KJ/(KG)(K)    16.3625  16.3625
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M, (1/n)            6.782    6.782
(dLV/dLP)t       -1.00002 -1.00000
(dLV/dLT)p         0.9998   0.9999
Cp, KJ/(KG)(K)     4.2665   4.1977
GAMMAs             1.4030   1.4125
SON VEL,M/SEC       716.1    654.7
MACH NUMBER         0.000    1.000

PERFORMANCE PARAMETERS

Ae/At                       1.0000
CSTAR, M/SEC                 881.2
CF                          0.7430
Ivac, M/SEC                 1118.2
Isp, M/SEC                   654.7


MOLE FRACTIONS

HO2               0.00007  0.00002
*H2               0.84104  0.84104
*O2               0.15889  0.15894

 * THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS
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             THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

          COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin =   999.3 PSIA
CASE = _______________

            REACTANT                    WT FRACTION      ENERGY      TEMP
                                         (SEE NOTE)     KJ/KG-MOL      K
FUEL        H2                           1.0000000        -0.000    298.150
OXIDANT     O2                           1.0000000        -0.000    298.150

O/F=    3.50000  %FUEL= 22.222222  R,EQ.RATIO= 2.267624  PHI,EQ.RATIO= 2.267624

                CHAMBER   THROAT
Pinf/P            1.0000   1.9008
P, BAR            68.901   36.248
T, K              298.12   247.56
RHO, KG/CU M    2.0661 1 1.3089 1
H, KJ/KG        -0.00000  -195.52
U, KJ/KG         -333.48  -472.45
G, KJ/KG        -4527.19 -3954.92
S, KJ/(KG)(K)    15.1858  15.1858

M, (1/n)           7.433    7.433
(dLV/dLP)t      -1.00002 -1.00000
(dLV/dLT)p        0.9998   0.9999
Cp, KJ/(KG)(K)    3.8947   3.8328
GAMMAs            1.4027   1.4121
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SON VEL,M/SEC      683.9    625.3
MACH NUMBER        0.000    1.000

PERFORMANCE PARAMETERS

Ae/At                      1.0000
CSTAR, M/SEC                841.8
CF                         0.7428
Ivac, M/SEC                1068.2
Isp, M/SEC                  625.3


MOLE FRACTIONS

HO2              0.00007  0.00002
*H2              0.81933  0.81934
*O2              0.18059  0.18064

 * THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS
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             THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

          COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin =   999.3 PSIA
CASE = _______________

            REACTANT                    WT FRACTION      ENERGY      TEMP
                                         (SEE NOTE)     KJ/KG-MOL      K
FUEL        H2                           1.0000000        -0.000    298.150
OXIDANT     O2                           1.0000000        -0.000    298.150

O/F=    4.00000  %FUEL= 20.000000  R,EQ.RATIO= 1.984171  PHI,EQ.RATIO= 1.984171

                CHAMBER   THROAT
Pinf/P            1.0000   1.9006
P, BAR            68.901   36.253
T, K              298.12   247.59
RHO, KG/CU M    2.2380 1 1.4178 1
H, KJ/KG        -0.00000  -180.48
U, KJ/KG         -307.87  -436.18
G, KJ/KG        -4245.60 -3706.53
S, KJ/(KG)(K)    14.2414  14.2414

M, (1/n)           8.051    8.051
(dLV/dLP)t      -1.00002 -1.00001
(dLV/dLT)p        0.9998   0.9999
Cp, KJ/(KG)(K)    3.5972   3.5409
GAMMAs            1.4024   1.4117
SON VEL,M/SEC      657.1    600.8
MACH NUMBER        0.000    1.000

PERFORMANCE PARAMETERS
```

Ae/At	1.0000
CSTAR, M/SEC	808.9
CF	0.7427
Ivac, M/SEC	1026.4
Isp, M/SEC	600.8

MOLE FRACTIONS

HO2	0.00008	0.00002
*H2	0.79872	0.79872
*O2	0.20120	0.20126

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 999.3 PSIA
CASE = _______________

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H2	1.0000000	-0.000	298.150
OXIDANT	O2	1.0000000	-0.000	298.150

O/F= 4.50000 %FUEL= 18.181818 R,EQ.RATIO= 1.763707 PHI,EQ.RATIO= 1.763707

	CHAMBER	THROAT
Pinf/P	1.0000	1.9003
P, BAR	68.901	36.257
T, K	298.11	247.62
RHO, KG/CU M	2.4014 1	1.5213 1
H, KJ/KG	-0.00000	-168.17
U, KJ/KG	-286.92	-406.51
G, KJ/KG	-4014.50	-3502.73
S, KJ/(KG)(K)	13.4663	13.4663
M, (1/n)	8.639	8.639
(dLV/dLP)t	-1.00002	-1.00001
(dLV/dLT)p	0.9998	0.9999
Cp, KJ/(KG)(K)	3.3539	3.3021
GAMMAs	1.4022	1.4113
SON VEL,M/SEC	634.3	580.0
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	780.9
CF	0.7426
Ivac, M/SEC	990.9
Isp, M/SEC	580.0

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MOLE FRACTIONS

HO2              0.00009  0.00002
*H2              0.77911  0.77912
*O2              0.22080  0.22086

 * THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS






             THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

          COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin =   999.3 PSIA
CASE = _______________

            REACTANT                    WT FRACTION      ENERGY      TEMP
                                         (SEE NOTE)     KJ/KG-MOL      K
FUEL        H2                           1.0000000        -0.000    298.150
OXIDANT     O2                           1.0000000        -0.000    298.150

O/F=    5.00000  %FUEL= 16.666667  R,EQ.RATIO= 1.587337  PHI,EQ.RATIO= 1.587337

                CHAMBER   THROAT
Pinf/P            1.0000   1.9001
P, BAR            68.901   36.262
T, K              298.11   247.65
RHO, KG/CU M    2.5570 1 1.6198 1
H, KJ/KG        -0.00000  -157.92
U, KJ/KG         -269.46  -381.78
G, KJ/KG        -3821.36 -3332.44
S, KJ/(KG)(K)    12.8186  12.8186

M, (1/n)           9.198    9.198
(dLV/dLP)t      -1.00002 -1.00001
(dLV/dLT)p        0.9998   0.9999
Cp, KJ/(KG)(K)    3.1511   3.1031
GAMMAs            1.4019   1.4109
SON VEL,M/SEC      614.6    562.0
MACH NUMBER        0.000    1.000

PERFORMANCE PARAMETERS

Ae/At                      1.0000
CSTAR, M/SEC                756.9
CF                         0.7425
Ivac, M/SEC                 960.3
Isp, M/SEC                  562.0


MOLE FRACTIONS

HO2              0.00009  0.00002
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*H2              0.76045  0.76046
*O2              0.23946  0.23952

 * THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS
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             THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

          COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin =   999.3 PSIA
CASE = _______________

            REACTANT                    WT FRACTION      ENERGY      TEMP
                                         (SEE NOTE)     KJ/KG-MOL      K
FUEL        H2                           1.0000000        -0.000    298.150
OXIDANT     O2                           1.0000000        -0.000    298.150

O/F=    5.50000  %FUEL= 15.384615  R,EQ.RATIO= 1.443033  PHI,EQ.RATIO= 1.443033

                CHAMBER   THROAT
Pinf/P            1.0000   1.8999
P, BAR            68.901   36.266
T, K              298.11   247.68
RHO, KG/CU M    2.7053 1 1.7138 1
H, KJ/KG        -0.00000  -149.25
U, KJ/KG         -254.69  -360.86
G, KJ/KG        -3657.47 -3187.98
S, KJ/(KG)(K)    12.2689  12.2689

M, (1/n)           9.732    9.732
(dLV/dLP)t      -1.00002 -1.00001
(dLV/dLT)p        0.9998   0.9999
Cp, KJ/(KG)(K)    2.9794   2.9347
GAMMAs            1.4017   1.4106
SON VEL,M/SEC      597.5    546.4
MACH NUMBER        0.000    1.000

PERFORMANCE PARAMETERS

Ae/At                      1.0000
CSTAR, M/SEC                735.9
CF                         0.7424
Ivac, M/SEC                 933.7
Isp, M/SEC                  546.3


MOLE FRACTIONS

HO2              0.00010  0.00003
*H2              0.74266  0.74267
*O2              0.25724  0.25731

 * THERMODYNAMIC PROPERTIES FITTED TO 20000.K
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NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

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Pin =   999.3 PSIA
CASE = _______________

            REACTANT                    WT FRACTION      ENERGY      TEMP
                                         (SEE NOTE)     KJ/KG-MOL      K
FUEL        H2                           1.0000000        -0.000    298.150
OXIDANT     O2                           1.0000000        -0.000    298.150

O/F=    6.00000  %FUEL= 14.285714  R,EQ.RATIO= 1.322780  PHI,EQ.RATIO= 1.322780

                 CHAMBER   THROAT
Pinf/P             1.0000   1.8997
P, BAR             68.901   36.270
T, K               298.11   247.70
RHO, KG/CU M    2.8468 1 1.8034 1
H, KJ/KG         -0.00000  -141.81
U, KJ/KG          -242.03  -342.93
G, KJ/KG         -3516.63 -3063.85
S, KJ/(KG)(K)     11.7966  11.7966

M, (1/n)           10.241   10.241
(dLV/dLP)t       -1.00003 -1.00001
(dLV/dLT)p         0.9997   0.9999
Cp, KJ/(KG)(K)     2.8323   2.7903
GAMMAs             1.4015   1.4103
SON VEL,M/SEC       582.4    532.6
MACH NUMBER         0.000    1.000

PERFORMANCE PARAMETERS

Ae/At                       1.0000
CSTAR, M/SEC                 717.4
CF                          0.7423
Ivac, M/SEC                  910.2
Isp, M/SEC                   532.6


MOLE FRACTIONS

HO2              0.00010  0.00003
*H2              0.72568  0.72569
*O2              0.27422  0.27428

 * THERMODYNAMIC PROPERTIES FITTED TO 20000.K
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NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

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             THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

          COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

 Pin =   999.3 PSIA
 CASE = _______________

             REACTANT                    WT FRACTION      ENERGY      TEMP
                                          (SEE NOTE)     KJ/KG-MOL      K
 FUEL        H2                           1.0000000        -0.000    298.150
 OXIDANT     O2                           1.0000000        -0.000    298.150

 O/F=    6.50000  %FUEL= 13.333333  R,EQ.RATIO= 1.221028  PHI,EQ.RATIO= 1.221028

                 CHAMBER   THROAT
 Pinf/P            1.0000   1.8995
 P, BAR            68.901   36.273
 T, K              298.10   247.73
 RHO, KG/CU M    2.9820 1 1.8891 1
 H, KJ/KG        -0.00000  -135.37
 U, KJ/KG         -231.06  -327.38
 G, KJ/KG        -3394.25 -2956.02
 S, KJ/(KG)(K)    11.3861  11.3861

 M, (1/n)          10.727   10.727
 (dLV/dLP)t      -1.00003 -1.00001
 (dLV/dLT)p        0.9997   0.9999
 Cp, KJ/(KG)(K)    2.7049   2.6652
 GAMMAs            1.4013   1.4100
 SON VEL,M/SEC      569.0    520.3
 MACH NUMBER        0.000    1.000

 PERFORMANCE PARAMETERS

 Ae/At                      1.0000
 CSTAR, M/SEC                701.0
 CF                         0.7423
 Ivac, M/SEC                 889.4
 Isp, M/SEC                  520.3


 MOLE FRACTIONS

 HO2              0.00011  0.00003
 *H2              0.70946  0.70947
 *O2              0.29043  0.29050

  * THERMODYNAMIC PROPERTIES FITTED TO 20000.K

 NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS
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             THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

          COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

 Pin =   999.3 PSIA
 CASE = _______________

             REACTANT                    WT FRACTION      ENERGY      TEMP
                                          (SEE NOTE)     KJ/KG-MOL      K
 FUEL        H2                           1.0000000        -0.000    298.150
 OXIDANT     O2                           1.0000000        -0.000    298.150

 O/F=    7.00000  %FUEL= 12.500000  R,EQ.RATIO= 1.133812  PHI,EQ.RATIO= 1.133812

                 CHAMBER   THROAT
 Pinf/P            1.0000   1.8993
 P, BAR            68.901   36.277
 T, K              298.10   247.75
 RHO, KG/CU M    3.1113 1 1.9710 1
 H, KJ/KG        -0.00000  -129.73
 U, KJ/KG         -221.45  -313.78
 G, KJ/KG        -3286.91 -2861.45
 S, KJ/(KG)(K)    11.0261  11.0261

 M, (1/n)          11.192   11.192
 (dLV/dLP)t      -1.00003 -1.00001
 (dLV/dLT)p        0.9997   0.9999
 Cp, KJ/(KG)(K)    2.5933   2.5558
 GAMMAs            1.4011   1.4097
 SON VEL,M/SEC      557.0    509.4
 MACH NUMBER        0.000    1.000

 PERFORMANCE PARAMETERS

 Ae/At                      1.0000
 CSTAR, M/SEC                686.3
 CF                         0.7422
 Ivac, M/SEC                 870.7
 Isp, M/SEC                  509.4


 MOLE FRACTIONS

 HO2              0.00011  0.00003
 *H2              0.69395  0.69396
 *O2              0.30594  0.30601

  * THERMODYNAMIC PROPERTIES FITTED TO 20000.K

 NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS
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             THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

          COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR
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```
Pin =   999.3 PSIA
CASE = _______________

            REACTANT                    WT FRACTION      ENERGY      TEMP
                                         (SEE NOTE)     KJ/KG-MOL      K
FUEL        H2                           1.0000000        -0.000    298.150
OXIDANT     O2                           1.0000000        -0.000    298.150

O/F=    7.50000  %FUEL= 11.764706  R,EQ.RATIO= 1.058224  PHI,EQ.RATIO= 1.058224

                CHAMBER   THROAT
Pinf/P            1.0000   1.8991
P, BAR            68.901   36.280
T, K              298.10   247.77
RHO, KG/CU M    3.2351 1 2.0493 1
H, KJ/KG        -0.00000  -124.75
U, KJ/KG         -212.98  -301.78
G, KJ/KG        -3191.96 -2777.82
S, KJ/(KG)(K)    10.7077  10.7077

M, (1/n)          11.637   11.637
(dLV/dLP)t      -1.00003 -1.00001
(dLV/dLT)p        0.9997   0.9999
Cp, KJ/(KG)(K)    2.4949   2.4592
GAMMAs            1.4009   1.4094
SON VEL,M/SEC      546.2    499.5
MACH NUMBER        0.000    1.000

PERFORMANCE PARAMETERS

Ae/At                      1.0000
CSTAR, M/SEC                673.1
CF                         0.7421
Ivac, M/SEC                 853.9
Isp, M/SEC                  499.5


MOLE FRACTIONS

HO2              0.00012  0.00003
*H2              0.67910  0.67912
*O2              0.32078  0.32085

 * THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS
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             THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

          COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin =   999.3 PSIA
CASE = _______________

            REACTANT                    WT FRACTION      ENERGY      TEMP
                                         (SEE NOTE)     KJ/KG-MOL      K
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FUEL        H2                          1.0000000       -0.000    298.150
OXIDANT     O2                          1.0000000       -0.000    298.150

O/F=    8.00000  %FUEL= 11.111111  R,EQ.RATIO= 0.992085  PHI,EQ.RATIO= 0.992085

                CHAMBER   THROAT
Pinf/P           1.0000   1.8990
P, BAR           68.901   36.283
T, K             298.10   247.79
RHO, KG/CU M    3.3536 1 2.1245 1
H, KJ/KG        -0.00000  -120.33
U, KJ/KG         -205.45  -291.12
G, KJ/KG        -3107.38 -2703.33
S, KJ/(KG)(K)    10.4240  10.4240

M, (1/n)          12.064   12.063
(dLV/dLP)t      -1.00003 -1.00001
(dLV/dLT)p        0.9997   0.9999
Cp, KJ/(KG)(K)    2.4074   2.3733
GAMMAs            1.4007   1.4091
SON VEL,M/SEC      536.5    490.6
MACH NUMBER        0.000    1.000

PERFORMANCE PARAMETERS

Ae/At                      1.0000
CSTAR, M/SEC                661.1
CF                         0.7420
Ivac, M/SEC                 838.7
Isp, M/SEC                  490.6


MOLE FRACTIONS

HO2             0.00012  0.00003
*H2             0.66488  0.66489
*O2             0.33500  0.33507

 * THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS
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             THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

          COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin =   999.3 PSIA
CASE = _______________

            REACTANT                    WT FRACTION      ENERGY      TEMP
                                         (SEE NOTE)     KJ/KG-MOL      K
FUEL        H2                          1.0000000       -0.000    298.150
OXIDANT     O2                          1.0000000       -0.000    298.150

O/F=    8.50000  %FUEL= 10.526316  R,EQ.RATIO= 0.933727  PHI,EQ.RATIO= 0.933727
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                CHAMBER   THROAT
Pinf/P            1.0000   1.8988
P, BAR            68.901   36.286
T, K              298.10   247.82
RHO, KG/CU M    3.4674 1 2.1965 1
H, KJ/KG        -0.00000  -116.37
U, KJ/KG         -198.71  -281.57
G, KJ/KG        -3031.52 -2636.54
S, KJ/(KG)(K)    10.1696  10.1696

M, (1/n)          12.473   12.472
(dLV/dLP)t      -1.00003 -1.00001
(dLV/dLT)p        0.9997   0.9999
Cp, KJ/(KG)(K)    2.3291   2.2965
GAMMAs            1.4006   1.4089
SON VEL,M/SEC      527.6    482.4
MACH NUMBER        0.000    1.000

PERFORMANCE PARAMETERS

Ae/At                      1.0000
CSTAR, M/SEC                650.2
CF                         0.7420
Ivac, M/SEC                 824.9
Isp, M/SEC                  482.4


MOLE FRACTIONS

HO2              0.00013  0.00003
*H2              0.65124  0.65125
*O2              0.34864  0.34871

 * THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS
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             THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

          COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin =   999.3 PSIA
CASE = _______________

            REACTANT                    WT FRACTION      ENERGY      TEMP
                                         (SEE NOTE)     KJ/KG-MOL      K
FUEL        H2                           1.0000000        -0.000    298.150
OXIDANT     O2                           1.0000000        -0.000    298.150

O/F=    9.00000  %FUEL= 10.000000  R,EQ.RATIO= 0.881854  PHI,EQ.RATIO= 0.881854

                CHAMBER   THROAT
Pinf/P            1.0000   1.8987
P, BAR            68.901   36.289
T, K              298.10   247.83
RHO, KG/CU M    3.5765 1 2.2656 1
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H, KJ/KG        -0.00000  -112.81
U, KJ/KG         -192.65  -272.98
G, KJ/KG        -2963.11 -2576.31
S, KJ/(KG)(K)     9.9401   9.9401

M, (1/n)          12.866   12.865
(dLV/dLP)t      -1.00003 -1.00001
(dLV/dLT)p        0.9997   0.9999
Cp, KJ/(KG)(K)    2.2586   2.2274
GAMMAs            1.4004   1.4086
SON VEL,M/SEC      519.4    475.0
MACH NUMBER        0.000    1.000

PERFORMANCE PARAMETERS

Ae/At                      1.0000
CSTAR, M/SEC                640.2
CF                         0.7419
Ivac, M/SEC                 812.2
Isp, M/SEC                  475.0


MOLE FRACTIONS

HO2              0.00013  0.00003
*H2              0.63814  0.63816
*O2              0.36173  0.36180

 * THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS
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             THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

          COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin =   999.3 PSIA
CASE = _______________

            REACTANT                    WT FRACTION      ENERGY      TEMP
                                         (SEE NOTE)     KJ/KG-MOL      K
FUEL        H2                           1.0000000        -0.000    298.150
OXIDANT     O2                           1.0000000        -0.000    298.150

O/F=    9.50000  %FUEL=  9.523810  R,EQ.RATIO= 0.835440  PHI,EQ.RATIO= 0.835440

                 CHAMBER   THROAT
Pinf/P            1.0000   1.8985
P, BAR            68.901   36.292
T, K              298.10   247.85
RHO, KG/CU M    3.6814 1 2.3320 1
H, KJ/KG        -0.00000  -109.59
U, KJ/KG         -187.16  -265.21
G, KJ/KG        -2901.08 -2521.71
S, KJ/(KG)(K)     9.7320   9.7320
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M, (1/n)          13.243   13.242
(dLV/dLP)t      -1.00003 -1.00001
(dLV/dLT)p        0.9997   0.9999
Cp, KJ/(KG)(K)    2.1949   2.1648
GAMMAs            1.4003   1.4084
SON VEL,M/SEC      511.9    468.2
MACH NUMBER        0.000    1.000

PERFORMANCE PARAMETERS

Ae/At                      1.0000
CSTAR, M/SEC                631.1
CF                         0.7418
Ivac, M/SEC                 800.6
Isp, M/SEC                  468.2


MOLE FRACTIONS

HO2             0.00013  0.00004
*H2             0.62557  0.62559
*O2             0.37430  0.37438

 * THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS
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             THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

          COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin =   999.3 PSIA
CASE = _______________

            REACTANT                    WT FRACTION      ENERGY      TEMP
                                         (SEE NOTE)     KJ/KG-MOL      K
FUEL        H2                           1.0000000        -0.000    298.150
OXIDANT     O2                           1.0000000        -0.000    298.150

O/F=   10.00000  %FUEL=  9.090909  R,EQ.RATIO= 0.793668  PHI,EQ.RATIO= 0.793668

                 CHAMBER   THROAT
Pinf/P            1.0000   1.8984
P, BAR            68.901   36.295
T, K              298.09   247.87
RHO, KG/CU M    3.7821 1 2.3959 1
H, KJ/KG        -0.00000  -106.66
U, KJ/KG         -182.17  -258.15
G, KJ/KG        -2844.57 -2471.98
S, KJ/(KG)(K)     9.5425   9.5425

M, (1/n)          13.605   13.604
(dLV/dLP)t      -1.00003 -1.00001
(dLV/dLT)p        0.9997   0.9999
Cp, KJ/(KG)(K)    2.1369   2.1079
GAMMAs            1.4001   1.4082
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SON VEL,M/SEC	505.0	461.9
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.00000
CSTAR, M/SEC	622.7
CF	0.7418
Ivac, M/SEC	789.9
Isp, M/SEC	461.9

MOLE FRACTIONS

HO2	0.00014	0.00004
*H2	0.61348	0.61350
*O2	0.38639	0.38647

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS